DS	**Session**	Tree 2	**Question Information**	Level 1 Challenge 72

Problem Description

You are given an N-dimensional array A. You can execute an operation that removes the greatest and smallest elements from the array and replaces them with their difference. As a result, the array's size will drop by one after each operation. You are assigned Q jobs, each of which contains an integer K. After K operations, you must provide the total of all the items in the array for each task.

Constraints:

$2 <= N <= 10^5$

$1 <= Q <= 10^5$

$0 <= A[i] <= 10^9$

$0 <= K < N$

Input:

First line contains two space-separated integers **N** and **Q**, denoting the number of elements in array and number of queries respectively.

Next line contains **N** space-separated integers denoting elements of the array.

Next **Q** lines contain a single integer **K**.

Output:

For each task, print answer in a new line.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 2
3 2 1 5 4
1
2
```

EXPECTED OUTPUT

```
13
9
```

Test Case 2

INPUT (STDIN)

```
5 3
13 12 11 15 14
1
2
3
```

EXPECTED OUTPUT

```
43
35
15
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
for(i=0;i<n;i++)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
4
```

Test Case 2

TOKEN COUNT

```
300
```

Test Case 3

NLOC

```
38
```

Code

k_array_diff_ops.c

```c
#include <stdio.h>
#include <stdlib.h>

int cmp_asc(const void* a, const void* b) {
    long long x = *(const long long*)a;
    long long y = *(const long long*)b;
    if (x < y) return -1;
    if (x > y) return 1;
    return 0;
}

int main() {
    int n, q;
    if (scanf("%d %d", &n, &q) != 2) return 0;

    long long* a = (long long*)malloc(n * sizeof(long long));
    long long total_sum = 0;
    int i;
    for(i=0;i<n;i++) {
        scanf("%lld", &a[i]);
        total_sum += a[i];
    }

    long long* ans = (long long*)malloc((n + 1) * sizeof(long long));
    ans[0] = total_sum;

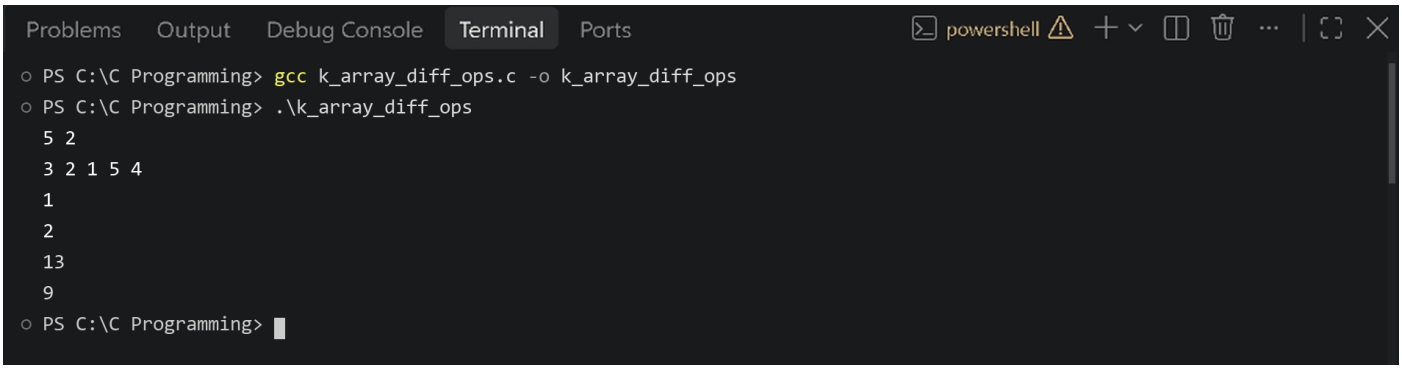
    int sz = n;
    long long cur_sum = total_sum;
    for (int step = 1; step < n; step++) {
        qsort(a, sz, sizeof(long long), cmp_asc);
        long long min_val = a[0];
        long long max_val = a[sz - 1];
        long long diff = max_val - min_val;
        cur_sum -= (2 * min_val);
        ans[step] = cur_sum;

        a[0] = diff;
        sz--;
    }

    for (int j = 0; j < q; j++) {
        int k;
        scanf("%d", &k);
        if (k < n) {
            printf("%lld\n", ans[k]);
        } else {
            printf("%lld\n", ans[n - 1]);
        }
    }

    free(a);
    free(ans);
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc k_array_diff_ops.c -o k_array_diff_ops
PS C:\C Programming> .\k_array_diff_ops
5 2
3 2 1 5 4
1
2
13
9
PS C:\C Programming>